

DEPLOYING MACHINE LEARNING MODELS IN PRODUCTION

02 ungraded lab tensorflow serving with docker C4 W1 Lab 2 TFS Docker

Source: 02_ungraded-lab-tensorflow-serving-with-docker_C4_W1_Lab_2_TFS_Docker.md

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property="og:image:width" content="1200" /><meta property="og:image:height" content="600"
/><meta property="og:site_name" content="GitHub" /><meta property="og:type"
content="object" /><meta property="og:title" content="machine-learning-engineering-for-
production-public/course4/week1-ungraded-labs/C4_W1_Lab_2_TFS_Docker.md at main · https-
deeplearning-ai/machine-learning-engineering-for-production-public" /><meta
property="og:url" content="https://github.com/https-deeplearning-ai/machine-learning-
engineering-for-production-public" /><meta property="og:description" content="Public repo
for DeepLearning.AI MLEP Specialization - machine-learning-engineering-for-production-
public/course4/week1-ungraded-labs/C4_W1_Lab_2_TFS_Docker.md at main · https-deeplearning-
ai/machin..." />

  <meta name="hostname" content="github.com">

  <meta name="expected-hostname" content="github.com">

```



```
<meta name="enabled-features"
content="TURBO_EXPERIMENT_RISKY, IMAGE_METRIC_TRACKING, GEOJSON_AZURE_MAPS, SUDO_MODAL_FOCUS">
```

```
<meta data-hydrostats="publish">
```

```
<link rel="canonical" href="https://github.com/https-deeplearning-ai/machine-learning-
engineering-for-production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_2_TFS_Docker.md" data-turbo-transient>
```

```
<div class="position-relative js-header-wrapper ">
  <a href="#start-of-content" class="px-2 py-4 color-bg-accent-emphasis color-fg-on-
emphasis show-on-focus js-skip-to-content">Skip to content</a>
  <span data-view-component="true" class="progress-pjax-loader Progress position-fixed
width-full">
<span style="width: 0%;" data-view-component="true" class="Progress-item progress-pjax-
loader-bar left-0 top-0 color-bg-accent-emphasis"></span>
```

```
<script crossorigin="anonymous" defer="defer" type="application/javascript"
src="https://github.githubassets.com/assets/vendors-node_modules_github_remote-
form_dist_index_js-node_modules_github_memoize_dist_esm_in-687f35-d131f0b6de8e.js">
</script>
```

Toggle navigation



```

<div class="flex-1">
  <a href="/signup?
ref_cta=Sign+up&ref_loc=header+logged+out&ref_page=%2F%3Cuser-name%3E%2F%3Crepo-
name%3E%2Fblob%2Fshow&source=header-repo"
    class="d-inline-block d-lg-none flex-order-1 f5 no-underline border color-border-
default rounded-2 px-2 py-1 color-fg-inherit"
    data-hydro-click="{"event_type":"authentication.click","payload":"
{"location_in_page":"site
header","repository_id":"null","auth_type":"SIGN_UP","originating_url":"https://github.com/https-deeplearning-ai/machine-learning-
engineering-for-production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_2_TFS_Docker.md","user_id":"null"}" data-hydro-click-
hmac="6cb67cfaccfa03b972bae243ce2b2cbae0e647f6a02730abef618d8431f7d794"
    >
    Sign up
  </a>
</div>

<div class="flex-1 flex-order-2 text-right">
  <button aria-label="Toggle navigation" aria-expanded="false" type="button" data-view-
component="true" class="js-details-target Button--link Button--medium Button d-lg-none
color-fg-inherit p-1">    <span class="Button-content">
    <span class="Button-label"><div class="HeaderMenu-toggle-bar rounded my-1"></div>
    <div class="HeaderMenu-toggle-bar rounded my-1"></div>
    <div class="HeaderMenu-toggle-bar rounded my-1"></div></span>
</span>
</div>

```

```

<div class="HeaderMenu--logged-out p-responsive height-fit position-lg-relative d-lg-flex
flex-column flex-auto pt-7 pb-4 top-0">
  <div class="header-menu-wrapper d-flex flex-column flex-self-end flex-lg-row flex-justify-
between flex-auto p-3 p-lg-0 rounded rounded-lg-0 mt-3 mt-lg-0">
    <nav class="mt-0 px-3 px-lg-0 mb-3 mb-lg-0" aria-label="Global">
      <ul class="d-lg-flex list-style-none">
        <li class="HeaderMenu-item position-relative flex-wrap flex-justify-between flex-
items-center d-block d-lg-flex flex-lg-nowrap flex-lg-items-center js-details-container js-
header-menu-item">
          <button type="button" class="HeaderMenu-link border-0 width-full width-lg-auto px-0 px-
lg-2 py-3 py-lg-2 no-wrap d-flex flex-items-center flex-justify-between js-details-target"
aria-expanded="false">
            Product
            <svg opacity="0.5" aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-chevron-down HeaderMenu-icon
ml-1">
<path d="M12.78 5.22a.749.749 0 0 1 0 1.06l-4.25 4.25a.749.749 0 0 1-1.06 0L3.22
6.28a.749.749 0 1 1 1.06-1.06L8.93 13.72-3.719a.749.749 0 0 1 1.06 0Z"></path>

```



Actions

Automate any workflow



[Packages](#)

[Host and manage packages](#)



[Security](#)

[Find and fix vulnerabilities](#)



[Codespaces](#)

[Instant dev environments](#)



[Copilot](#)

[Write better code with AI](#)



[Code review](#)

[Manage code changes](#)



[Issues](#)

[Plan and track work](#)



Discussions

Collaborate outside of code

```

    </ul>
  </div>
  <div class="px-lg-4">
    <span class="d-block h4 color-fg-default my-1" id="product-explore-
heading">Explore</span>
    <ul class="list-style-none f5" aria-labelledby="product-explore-heading">
      <li>

```

All features

```
<li>
```

Documentation

```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-
view-component="true" class="octicon octicon-link-external HeaderMenu-external-icon
color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0 1
12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-1h4.146a.25.25
0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751 0 0 1-
1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604 12">
</path>

```

```
<li>
```

GitHub Skills

```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-
view-component="true" class="octicon octicon-link-external HeaderMenu-external-icon
color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0 1
12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-1h4.146a.25.25
0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751 0 0 1-
1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604 12">
</path>

```

```
<li>
```

[Blog](#)

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0 1
12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-1h4.146a.25.25
0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751 0 0 1-
1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604 12">
</path>
```

```
</ul>
</div>
</div>
```

```
<li class="HeaderMenu-item position-relative flex-wrap flex-justify-between
flex-items-center d-block d-lg-flex flex-lg-nowrap flex-lg-items-center js-details-container
js-header-menu-item">
  <button type="button" class="HeaderMenu-link border-0 width-full width-lg-auto px-0
px-lg-2 py-3 py-lg-2 no-wrap d-flex flex-items-center flex-justify-between js-details-
target" aria-expanded="false">
    Solutions
    <svg opacity="0.5" aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-chevron-down HeaderMenu-
icon ml-1">
<path d="M12.78 5.22a.749.749 0 0 1 0 1.06l-4.25 4.25a.749.749 0 0 1-1.06 0L3.22
6.28a.749.749 0 1 1 1.06-1.06L8 8.939l3.72-3.719a.749.749 0 0 1 1.06 0Z"></path>
```

For

- [Enterprise](#)

```
<li>
```

[Teams](#)

```
<li>
```

[Startups](#)

```
<li>
```

Education

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-
icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0
1 12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-
1h4.146a.25.25 0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751
0 0 1-1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604
1Z"></path>
```

```
</ul>
</div>
<div class="border-bottom pb-3 mb-3">
  <span class="d-block h4 color-fg-default my-1" id="solutions-by-solution-
heading">By Solution</span>
  <ul class="list-style-none f5" aria-labelledby="solutions-by-solution-
heading">
    <li>
```

CI/CD & Automation

```
<li>
```

DevOps

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-
icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0
1 12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-
1h4.146a.25.25 0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751
0 0 1-1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604
1Z"></path>
```

```
<li>
```

DevSecOps

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-
icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0
1 12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-
1h4.146a.25.25 0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751
0 0 1-1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604
1Z"></path>
```

```
</ul>
</div>
<div class="">
  <span class="d-block h4 color-fg-default my-1" id="solutions-case-studies-
heading">Case Studies</span>
  <ul class="list-style-none f5" aria-labelledby="solutions-case-studies-
heading">
    <li>
```

Customer Stories

```
<li>
```

Resources

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-
icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0
1 12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-
1h4.146a.25.25 0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751
0 0 1-1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604
1Z"></path>
```

```
</ul>
</div>
</div>
```

```

<li class="HeaderMenu-item position-relative flex-wrap flex-justify-between
flex-items-center d-block d-lg-flex flex-lg-nowrap flex-lg-items-center js-details-
container js-header-menu-item">
  <button type="button" class="HeaderMenu-link border-0 width-full width-lg-auto px-0
px-lg-2 py-3 py-lg-2 no-wrap d-flex flex-items-center flex-justify-between js-details-
target" aria-expanded="false">
    Open Source
    <svg opacity="0.5" aria-hidden="true" height="16" viewBox="0 0 16 16"
version="1.1" width="16" data-view-component="true" class="octicon octicon-chevron-
down HeaderMenu-icon ml-1">
<path d="M12.78 5.22a.749.749 0 0 1 0 1.06l-4.25 4.25a.749.749 0 0 1-1.06 0L3.22
6.28a.749.749 0 1 1 1.06-1.06L8.939 3.72-3.719a.749.749 0 0 1 1.06 0Z"></path>

```

```

■ <div>
  <div class="color-fg-default h4">GitHub Sponsors</div>
  Fund open source developers
</div>

```

```

  </ul>
</div>
<div class="border-bottom pb-3 mb-3">
  <ul class="list-style-none f5" >
    <li>

```

```

<div>
  <div class="color-fg-default h4">The ReadME Project</div>
  GitHub community articles
</div>

```

```

  </ul>
</div>
<div class="">
  <span class="d-block h4 color-fg-default my-1" id="open-source-
repositories-heading">Repositories</span>
  <ul class="list-style-none f5" aria-labelledby="open-source-repositories-
heading">
    <li>

```

Topics

```
<li>
```

Trending

```
<li>
```

Collections

```

    </ul>
  </div>
</div>

```

```

    <li class="HeaderMenu-item position-relative flex-wrap flex-justify-
between flex-items-center d-block d-lg-flex flex-lg-nowrap flex-lg-items-center js-
details-container js-header-menu-item">
<a class="HeaderMenu-link no-underline px-0 px-lg-2 py-3 py-lg-2 d-block d-lg-
inline-block" data-analytics-event="{&quot;category&quot;:&quot;Header menu top
item (logged out)&quot;;&quot;action&quot;:&quot;click to go to
Pricing&quot;;&quot;label&quot;:&quot;ref_cta:Pricing&quot;;}"
href="/pricing">Pricing</a>

```

```

    </ul>
  </nav>

<div class="d-lg-flex flex-items-center px-3 px-lg-0 mb-3 mb-lg-0 text-center
text-lg-left">
  <div class="d-lg-flex min-width-0 mb-2 mb-lg-0">

```

```

placeholder="Search" data-unscoped-placeholder="Search
GitHub" data-scoped-placeholder="Search" autocapitalize="off" role="combobox" aria-
haspopup="listbox" aria-expanded="false" aria-autocomplete="list" aria-
controls="jump-to-results" aria-label="Search" data-jump-to-suggestions-
path="/_graphql/GetSuggestedNavigationDestinations" spellcheck="false"
autocomplete="off" > <input type="hidden" data-csrf="true" class="js-data-jump-to-
suggestions-path-csrf"
value="jD797szkDR78exHA3U/s/Z+Uav9Sb2TNdPGX01cXFxaFvIMgGInCMcPhwVko4L0LQ6pw1SY4Tm7k
eSWK7nt0Kw==" /> <input type="hidden" class="js-site-search-type-field" name="type"
> <svg xmlns="http://www.w3.org/2000/svg" width="22" height="20" aria-hidden="true"
class="mr-1 header-search-key-slash"><path fill="none" stroke="#979A9C" opacity=".4"
d="M3.5.5h12c1.7 0 3 1.3 3 3v13c0 1.7-1.3 3-3 3h-12c-1.7 0-3-1.3-3-3v-13c0-1.7 1.3-
3 3-3z"></path><path fill="#979A9C" d="M11.8 6L8 15.1h-.9L10.8 6h1z"></path></svg>
<div class="Box position-absolute overflow-hidden d-none jump-to-suggestions js-
jump-to-suggestions-container">

```




```

<img class="avatar mr-2 flex-shrink-0 js-jump-to-suggestion-avatar d-none"
alt="" aria-label="Team" src="" width="28" height="28">

<div class="jump-to-suggestion-name js-jump-to-suggestion-name flex-auto
overflow-hidden text-left no-wrap css-truncate css-truncate-target">
</div>

<div class="border rounded-2 flex-shrink-0 color-bg-subtle px-1 color-fg-muted
ml-1 f6 d-none js-jump-to-badge-search">
  <span class="js-jump-to-badge-search-text-default d-none" aria-label="in
this repository">
    In this repository
  </span>
  <span class="js-jump-to-badge-search-text-global d-none" aria-label="in all
of GitHub">
    All GitHub
  </span>
  <span aria-hidden="true" class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

<div aria-hidden="true" class="border rounded-2 flex-shrink-0 color-bg-subtle
px-1 color-fg-muted ml-1 f6 d-none d-on-nav-focus js-jump-to-badge-jump">
  Jump to
  <span class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

```

- No suggested jump to results
- 🗨️ 📄 🔍

```

<img class="avatar mr-2 flex-shrink-0 js-jump-to-suggestion-avatar d-none"
alt="" aria-label="Team" src="" width="28" height="28">

<div class="jump-to-suggestion-name js-jump-to-suggestion-name flex-auto
overflow-hidden text-left no-wrap css-truncate css-truncate-target">
</div>

<div class="border rounded-2 flex-shrink-0 color-bg-subtle px-1 color-fg-muted
ml-1 f6 d-none js-jump-to-badge-search">
  <span class="js-jump-to-badge-search-text-default d-none" aria-label="in
this repository">
    In this repository
  </span>
  <span class="js-jump-to-badge-search-text-global d-none" aria-label="in all
of GitHub">
    All GitHub
  </span>
  <span aria-hidden="true" class="d-inline-block ml-1 v-align-middle">↵</span>
</div>

<div aria-hidden="true" class="border rounded-2 flex-shrink-0 color-bg-subtle
px-1 color-fg-muted ml-1 f6 d-none d-on-nav-focus js-jump-to-badge-jump">
  Jump to
  <span class="d-inline-block ml-1 v-align-middle">↵</span>
</div>

```



```

<img class="avatar mr-2 flex-shrink-0 js-jump-to-suggestion-avatar d-none"
alt="" aria-label="Team" src="" width="28" height="28">

<div class="jump-to-suggestion-name js-jump-to-suggestion-name flex-auto
overflow-hidden text-left no-wrap css-truncate css-truncate-target">
</div>

<div class="border rounded-2 flex-shrink-0 color-bg-subtle px-1 color-fg-muted
ml-1 f6 d-none js-jump-to-badge-search">
  <span class="js-jump-to-badge-search-text-default d-none" aria-label="in
this organization">
    In this organization
  </span>
  <span class="js-jump-to-badge-search-text-global d-none" aria-label="in all
of GitHub">
    All GitHub
  </span>
  <span aria-hidden="true" class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

<div aria-hidden="true" class="border rounded-2 flex-shrink-0 color-bg-subtle
px-1 color-fg-muted ml-1 f6 d-none d-on-nav-focus js-jump-to-badge-jump">
  Jump to
  <span class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

```



```

<img class="avatar mr-2 flex-shrink-0 js-jump-to-suggestion-avatar d-none"
alt="" aria-label="Team" src="" width="28" height="28">

<div class="jump-to-suggestion-name js-jump-to-suggestion-name flex-auto
overflow-hidden text-left no-wrap css-truncate css-truncate-target">
</div>

<div class="border rounded-2 flex-shrink-0 color-bg-subtle px-1 color-fg-muted
ml-1 f6 d-none js-jump-to-badge-search">
  <span class="js-jump-to-badge-search-text-default d-none" aria-label="in
this repository">
    In this repository
  </span>
  <span class="js-jump-to-badge-search-text-global d-none" aria-label="in all
of GitHub">
    All GitHub
  </span>
  <span aria-hidden="true" class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

<div aria-hidden="true" class="border rounded-2 flex-shrink-0 color-bg-subtle
px-1 color-fg-muted ml-1 f6 d-none d-on-nav-focus js-jump-to-badge-jump">
  Jump to
  <span class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

```

```

</div>
</label>

```

```

</div>

<div class="position-relative mr-lg-3 d-lg-inline-block">
  <a href="/login?return_to=https%3A%2Fgithub.com%2Fhttps-deeplearning-ai%2Fmachine-learning-engineering-for-production-public%2Fblob%2Fmain%2Fcourse4%2Fweek1-ungraded-labs%2FC4_W1_Lab_2_TFS_Docker.md"
    class="HeaderMenu-link HeaderMenu-link--sign-in flex-shrink-0 no-underline d-block d-lg-inline-block border border-lg-0 rounded rounded-lg-0 p-2 p-lg-0"
    data-hydro-click="{&quot;event_type&quot;:&quot;authentication.click&quot;;&quot;payload&quot;:{&quot;location_in_page&quot;:&quot;site header menu&quot;;&quot;repository_id&quot;:null,&quot;auth_type&quot;:&quot;SIGN_UP&quot;;&quot;originating_url&quot;:&quot;https://github.com/https-deeplearning-ai/machine-learning-engineering-for-production-public/blob/main/course4/week1-ungraded-labs/C4_W1_Lab_2_TFS_Docker.md&quot;;&quot;user_id&quot;:null}}}" data-hydro-click-hmac="7c0e5fa8e235205668d5d9a0f72ba671de9d11c9924c0603469b632f681380cd"
    data-ga-click="(Logged out) Header, clicked Sign in, text:sign-in">
    Sign in
  </a>
</div>

  <a href="/signup?ref_cta=Sign+up&amp;ref_loc=header+logged+out&amp;ref_page=%2F%3Cuser-name%3E%2F%3Crepo-name%3E%2Fblob%2Fshow&amp;source=header-repo&amp;source_repo=https-deeplearning-ai%2Fmachine-learning-engineering-for-production-public"
    class="HeaderMenu-link HeaderMenu-link--sign-up flex-shrink-0 d-none d-lg-inline-block no-underline border color-border-default rounded px-2 py-1"
    data-hydro-click="{&quot;event_type&quot;:&quot;authentication.click&quot;;&quot;payload&quot;:{&quot;location_in_page&quot;:&quot;site header menu&quot;;&quot;repository_id&quot;:null,&quot;auth_type&quot;:&quot;SIGN_UP&quot;;&quot;originating_url&quot;:&quot;https://github.com/https-deeplearning-ai/machine-learning-engineering-for-production-public/blob/main/course4/week1-ungraded-labs/C4_W1_Lab_2_TFS_Docker.md&quot;;&quot;user_id&quot;:null}}}" data-hydro-click-hmac="7c0e5fa8e235205668d5d9a0f72ba671de9d11c9924c0603469b632f681380cd"
    data-analytics-event="{&quot;category&quot;:&quot;Sign up&quot;;&quot;action&quot;:&quot;click to sign up for account&quot;;&quot;label&quot;:&quot;ref_page:/&lt;user-name&gt;/&lt;repo-name&gt;/blob/show;ref_cta:Sign up;ref_loc:header logged out&quot;}">
    >
    Sign up
  </a>
</div>
</div>
</div>

```

```
<div hidden="hidden" data-view-component="true" class="js-stale-session-flash flash flash-warn mb-3">
```

```
  <svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-view-component="true" class="octicon octicon-alert">
    <path d="M6.457 1.047c.659-1.234 2.427-1.234 3.086 0l6.082 11.378A1.75 1.75 0 0 1 14.082 15H1.918a1.75 1.75 0 0 1-1.543-2.575Zm1.763.707a.25.25 0 0 0-.44 0L1.698 13.132a.25.25 0 0 0 .22.368h12.164a.25.25 0 0 0 .22-.368Zm.53 3.996v2.5a.75.75 0 0 1-1.5 0v-2.5a.75.75 0 0 1 1.5 0ZM9 11a1 1 0 1 1-2 0 1 1 0 0 1 2 0Z"></path>
```

```
<button class="flash-close js-flash-close" type="button" aria-label="Close">
  <svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-view-component="true" class="octicon octicon-x">
    <path d="M3.72 3.72a.75.75 0 0 1 1.06 0L8 6.94l3.22-3.22a.749.749 0 0 1 1.275.326.749.749 0 0 1-.215.734L9.06 8l3.22 3.22a.749.749 0 0 1-.326 1.275.749.749 0 0 1-.734-.215L8 9.06l-3.22 3.22a.751.751 0 0 1-1.042-.018.751.751 0 0 1-.018-1.042L6.94 8 3.72 4.78a.75.75 0 0 1 0-1.06Z"></path>
```

```
<div id="js-flash-container" data-turbo-replace>
```

```
<include-fragment class="js-notification-shelf-include-fragment" data-base-src="https://github.com/notifications/beta/shelf"></include-fragment>
```

```
<div class="d-flex flex-wrap flex-justify-end mb-3 px-3 px-md-4 px-lg-5" style="gap: 1rem;">
```

```
  <div class="flex-auto min-width-0 width-fit mr-3">
```



```
<span class="author flex-self-stretch" itemprop="author">
  <a class="url fn" rel="author" data-hovercard-type="organization" data-hovercard-url="/orgs/https-deeplearning-ai/hovercard" data-octo-click="hovercard-link-click" data-octo-dimensions="link_type:self" href="/https-deeplearning-ai">
    https-deeplearning-ai
```

/ [machine-learning-engineering-for-production-public](#)

```
<span></span><span class="Label Label--secondary v-align-middle mr-1">Public</span>
```

```
</div>
```

```
<div id="repository-details-container" data-turbo-replace>
  <ul class="pagehead-actions flex-shrink-0 d-none d-md-inline" style="padding: 2px 0;">
```

 [Notifications](#)

 [Fork 1.9k](#)

 [Star 1.5k](#) 

```
</div>
</div>
```

```
<div id="responsive-meta-container" data-turbo-replace>
```

```
<nav data-pjax="#js-repo-pjax-container" aria-label="Repository" data-view-
component="true" class="js-repo-nav js-sidenav-container-pjax js-responsive-underlinenav
overflow-hidden UnderlineNav px-3 px-md-4 px-lg-5">
```

- ```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-code UnderlineNav-octicon
d-none d-sm-inline">
<path d="m11.28 3.22 4.25 4.25a.75.75 0 0 1 0 1.06l-4.25 4.25a.749.749 0 0 1-
1.275-.326.749.749 0 0 1 .215-.734l13.94 8l-3.72-3.72a.749.749 0 0 1 .326-1.275.749.749
0 0 1 .734.215zm-6.56 0a.751.751 0 0 1 1.042.018.751.751 0 0 1 .018 1.042l2.06 8l3.72
3.72a.749.749 0 0 1-.326 1.275.749.749 0 0 1-.734-.215l.47 8.53a.75.75 0 0 1 0-1.06z">
</path>
```

## [Code](#)

- ```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-issue-opened UnderlineNav-
octicon d-none d-sm-inline">
<path d="M8 9.5a1.5 1.5 0 1 0 0-3 1.5 1.5 0 0 0 0 3z"></path><path d="M8 0a8 8 0 1 1 0
16A8 8 0 0 1 8 0zM1.5 8a6.5 6.5 0 1 0 13 0 6.5 6.5 0 0 0-13 0z"></path>
```

[Issues 18](#)

- ```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-git-pull-request
UnderlineNav-octicon d-none d-sm-inline">
<path d="M1.5 3.25a2.25 2.25 0 1 1 3 2.122v5.256a2.251 2.251 0 1 1-1.5 0V5.372A2.25 2.25
0 0 1 1.5 3.25zm5.677-.177L9.573.677A.25.25 0 0 1 10 .854V2.5h1A2.5 2.5 0 0 1 13.5
5v5.628a2.251 2.251 0 1 1-1.5 0V5a1 1 0 0 0-1-1h-1v1.646a.25.25 0 0 1-.427.177L7.177
3.427a.25.25 0 0 1 0-.354zM3.75 2.5a.75.75 0 1 0 0 1.5.75.75 0 0 0 0-1.5zm0 9.5a.75.75 0
1 0 0 1.5.75.75 0 0 0 0-1.5zm8.25.75a.75.75 0 1 0 1.5 0 .75.75 0 0 0-1.5 0z"></path>
```

## [Pull requests 2](#)

- ```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-play UnderlineNav-octicon
d-none d-sm-inline">
<path d="M8 0a8 8 0 1 1 0 16A8 8 0 0 1 8 0ZM1.5 8a6.5 6.5 0 1 0 13 0 6.5 6.5 0 0 0-13
0Zm4.879-2.773 4.264 2.559a.25.25 0 0 1 0 .428l-4.264 2.559A.25.25 0 0 1 6
10.559V5.442a.25.25 0 0 1 .379-.215Z"></path>

```

[Actions](#)

- ```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-table UnderlineNav-octicon
d-none d-sm-inline">
<path d="M0 1.75C0 .784.784 0 1.75 0h12.5C15.216 0 16 .784 16 1.75v12.5A1.75 1.75 0 0 1
14.25 16H1.75A1.75 1.75 0 0 1 0 14.25ZM6.5 6.5v8h7.75a.25.25 0 0 0 .25-.25V6.5Zm8-
1.5V1.75a.25.25 0 0 0-.25-.25H6.5V5Zm-13 1.5v7.75c0 .138.112.25.25.25H5v-8ZM5
5V1.5H1.75a.25.25 0 0 0-.25.25V5Z"></path>

```

### [Projects](#)

- ```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-shield UnderlineNav-octicon
d-none d-sm-inline">
<path d="M7.467.133a1.748 1.748 0 0 1 1.066 0l5.25 1.68A1.75 1.75 0 0 1 15 3.48V7c0
1.566-.32 3.182-1.303 4.682-.983 1.498-2.585 2.813-5.032 3.855a1.697 1.697 0 0 1-1.33
0c-2.447-1.042-4.049-2.357-5.032-3.855C1.32 10.182 1 8.566 1 7V3.48a1.75 1.75 0 0 1
1.217-1.667Zm.61 1.429a.25.25 0 0 0-.153 0l-5.25 1.68a.25.25 0 0 0-.174.238V7c0
1.358.275 2.666 1.057 3.86.784 1.194 2.121 2.34 4.366 3.297a.196.196 0 0 0 .154
0c2.245-.956 3.582-2.104 4.366-3.298C13.225 9.666 13.5 8.36 13.5 7V3.48a.251.251 0 0
0-.174-.237l-5.25-1.68ZM8.75 4.75v3a.75.75 0 0 1-1.5 0v-3a.75.75 0 0 1 1.5 0ZM9 10.5a1 1
0 1 1-2 0 1 1 0 0 1 2 0Z"></path>

```

[Security](#)

- ```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-graph UnderlineNav-octicon
d-none d-sm-inline">
<path d="M1.5 1.75V13.5h13.75a.75.75 0 0 1 0 1.5H.75a.75.75 0 0 1-.75-.75V1.75a.75.75 0
0 1 1.5 0Zm14.28 2.53-5.25 5.25a.75.75 0 0 1-1.06 0L7 7.06 4.28 9.78a.751.751 0 0 1-
1.042-.018.751.751 0 0 1-.018-1.042l3.25-3.25a.75.75 0 0 1 1.06 0L10 7.94l4.72-
4.72a.751.751 0 0 1 1.042.018.751.751 0 0 1 .018 1.042Z"></path>

```

### [Insights](#)

### [Permalink](#)

► 🔑 main



[View all tags](#)

## Name already in use

✕

```
<div data-view-component="true"> A tag already exists with the provided branch
name. Many Git commands accept both tag and branch names, so creating this branch may cause
unexpected behavior. Are you sure you want to create this branch?
```

```
</div>
<footer class="Overlay-footer Overlay-footer--alignEnd">
 <button data-close-dialog-id="warn-tag-match-create-branch-dialog" type="button"
data-view-component="true" class="btn"> Cancel
```

Create

## [machine-learning-engineering-for-production-public/course4/week1-ungraded-labs/C4\\_W1\\_Lab\\_2\\_TFS\\_Docker.md](#)

[Go to file](#)

► ...

```
<div id="spoof-warning" class="mt-0 pb-3" hidden aria-hidden>
```

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-view-
component="true" class="octicon octicon-alert float-left mt-1">
<path d="M6.457 1.047c.659-1.234 2.427-1.234 3.086 0l6.082 11.378A1.75 1.75 0 0 1 14.082
15H1.918a1.75 1.75 0 0 1-1.543-2.575Zm1.763.707a.25.25 0 0 0-.44 0L1.698 13.132a.25.25 0 0
0 .22.368h12.164a.25.25 0 0 0 .22-.368Zm.53 3.996v2.5a.75.75 0 0 1-1.5 0v-2.5a.75.75 0 0 1
1.5 0ZM9 11a1 1 0 1 1-2 0 1 1 0 0 1 2 0Z"></path>
```

```
<div class="overflow-hidden">This commit does not belong to any branch on this repository,
and may belong to a fork outside of the repository.</div>
```

```
<include-fragment src="/https-deeplearning-ai/machine-learning-engineering-for-production-
public/spoofed_commit_check/34af5af3f5b09a52a1537292b0fe14a29ae16b8d" data-test-
selector="spoofed-commit-check"></include-fragment>
```

```
<div class="Box d-flex flex-column flex-shrink-0 mb-3">
```



[andres-zartab](#)

```


 <a data-pjax="true" title="C4W1 lab2 - Fixed typos" class="Link--secondary"
href="/https-deeplearning-ai/machine-learning-engineering-for-production-
public/commit/9fa0038be3fb765156d0950b3473013737225d37">C4W1 lab2 - Fixed typos

 </div>

 <include-fragment accept="text/fragment+html" src="/https-deeplearning-ai/machine-
learning-engineering-for-production-
public/commit/9fa0038be3fb765156d0950b3473013737225d37/rollup?direction=e" class="d-
inline"></include-fragment>

</div>
<div class="ml-3 d-flex flex-shrink-0 flex-items-center flex-justify-end color-fg-muted no-
wrap">

 Latest commit
 <a class="text-small text-mono Link--secondary" href="/https-deeplearning-
ai/machine-learning-engineering-for-production-
public/commit/9fa0038be3fb765156d0950b3473013737225d37" data-pjax>9fa0038
 <relative-time datetime="2023-02-02T22:36:14Z"
class="no-wrap">Feb 2, 2023</relative-time>

 <a data-pjax href="/https-deeplearning-ai/machine-learning-engineering-for-
production-public/commits/main/course4/week1-ungraded-labs/C4_W1_Lab_2_TFS_Docker.md"
class="ml-3 no-wrap Link--primary no-underline">
 <svg text="gray" aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-history">
<path d="m.427 1.927 1.215 1.215a8.002 8.002 0 1 1-1.6 5.685.75.75 0 1 1 1.493-.154 6.5 6.5
0 1 0 1.18-4.458l1.358 1.358A.25.25 0 0 1 3.896 6H.25A.25.25 0 0 1 0 5.75V2.104a.25.25 0 0
1 .427-.177ZM7.75 4a.75.75 0 0 1 .75.75v2.992l2.028.812a.75.75 0 0 1-.557 1.392l-2.5-
1A.751.751 0 0 1 7 8.25v-3.5A.75.75 0 0 1 7.75 4Z"></path>

```

## History

► 1

```

 contributor
 </summary>
 <details-dialog
 class="Box Box--overlay d-flex flex-column anim-fade-in fast"
 aria-label="Users who have contributed to this file"
 src="/https-deeplearning-ai/machine-learning-engineering-for-production-
public/contributors-list/main/course4/week1-ungraded-labs/C4_W1_Lab_2_TFS_Docker.md"
 preload>
 <div class="Box-header">
 <button class="Box-btn-octicon btn-octicon float-right" type="button" aria-
label="Close dialog" data-close-dialog>
 <svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-x">
<path d="M3.72 3.72a.75.75 0 0 1 1.06 0L8 6.94l3.22-3.22a.749.749 0 0 1 1.275.326.749.749 0
0 1-.215.734L9.06 8l3.22 3.22a.749.749 0 0 1-.326 1.275.749.749 0 0 1-.734-.215L8 9.06l-
3.22 3.22a.751.751 0 0 1-1.042-.018.751.751 0 0 1-.018-1.042L6.94 8 3.72 4.78a.75.75 0 0 1
0-1.06Z"></path>

```

## Users who have contributed to this file



```
<readme-toc>
```

```
<div data-target="readme-toc.content" class="Box mt-3 position-relative">
```

```
<details
```

data-target="readme-toc.trigger" data-menu-hydro-

click="{\"event\_type\":\"repository\_toc\_menu.click\",\"payload\":

{\"target\":\"trigger\",\"repository\_id\":313476213,\"originating\_url\":\"[https://github.com/https-deeplearning-ai/machine-learning-engineering-for-production-public/blob/main/course4/week1-ungraded-labs/C4\\_W1\\_Lab\\_2\\_TFS\\_Docker.md](https://github.com/https-deeplearning-ai/machine-learning-engineering-for-production-public/blob/main/course4/week1-ungraded-labs/C4_W1_Lab_2_TFS_Docker.md)\";\";\"user\_id\";:null}}\" data-menu-hydro-click-

hmac=\"bbc4b32411f7f685b34b0a09afc78fe8a4d4af82913ba283b55e986ec7d525c1\" class=\"dropdown

details-reset details-overlay\"



```

<div class="SelectMenu-list SelectMenu-list--borderless p-2" style="overscroll-behavior:
contain;">
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 text-emphasized" style="-webkit-box-orient: vertical; padding-left: 12px;"
data-action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="
{"event_type":"repository_toc_menu.click";"payload":
{"target":"entry";"repository_id":313476213,"originating
_url":"https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_2_TFS_Docker.md";"user_id":null}}" data-hydro-click-
hmac="adf01ada59a0e9dbe6f513b9baadff3125575d23e2a5cb652ea1e9216476da21" href="#ungraded-
lab-first-look-at-tensorflow-serving-with-docker">Ungraded lab: First look at Tensorflow
Serving with Docker
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="
{"event_type":"repository_toc_menu.click";"payload":
{"target":"entry";"repository_id":313476213,"originating
_url":"https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_2_TFS_Docker.md";"user_id":null}}" data-hydro-click-
hmac="adf01ada59a0e9dbe6f513b9baadff3125575d23e2a5cb652ea1e9216476da21" href="#pulling-the-
image">Pulling the image
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="
{"event_type":"repository_toc_menu.click";"payload":
{"target":"entry";"repository_id":313476213,"originating
_url":"https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_2_TFS_Docker.md";"user_id":null}}" data-hydro-click-
hmac="adf01ada59a0e9dbe6f513b9baadff3125575d23e2a5cb652ea1e9216476da21" href="#clone-the-
repo-with-the-dummy-model">Clone the repo with the dummy model
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="
{"event_type":"repository_toc_menu.click";"payload":
{"target":"entry";"repository_id":313476213,"originating
_url":"https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_2_TFS_Docker.md";"user_id":null}}" data-hydro-click-
hmac="adf01ada59a0e9dbe6f513b9baadff3125575d23e2a5cb652ea1e9216476da21" href="#run-
tensorflow-serving">Run Tensorflow Serving
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="
{"event_type":"repository_toc_menu.click";"payload":
{"target":"entry";"repository_id":313476213,"originating
_url":"https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_2_TFS_Docker.md";"user_id":null}}" data-hydro-click-
hmac="adf01ada59a0e9dbe6f513b9baadff3125575d23e2a5cb652ea1e9216476da21" href="#consuming-
the-model">Consuming the model

```

```

 <a role="menuItem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="
{"event_type":"repository_toc_menu.click";"payload":
{"target":"entry";"repository_id":313476213,"originating
_url":"https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_2_TFS_Docker.md";"user_id":null}}" data-hydro-click-
hmac="adf01ada59a0e9dbe6f513b9baadff3125575d23e2a5cb652ea1e9216476da21" href="#stopping-
the-server">Stopping the server
 </div>
</div>

```

```

```

```
143 lines (89 sloc)
```

```

```

```
7.7 KB
```

```

<a class="dropdown-item" href="/https-deeplearning-ai/machine-learning-
engineering-for-production-public/blame/main/course4/week1-ungraded-
labs/C4_W1_Lab_2_TFS_Docker.md">

```

```
<div class="BtnGroup">
```

```

 <a data-permalink-href="/https-deeplearning-ai/machine-learning-engineering-for-
production-public/raw/34af5af3f5b09a52a1537292b0fe14a29ae16b8d/course4/week1-ungraded-
labs/C4_W1_Lab_2_TFS_Docker.md" href="/https-deeplearning-ai/machine-learning-engineering-
for-production-public/raw/main/course4/week1-ungraded-labs/C4_W1_Lab_2_TFS_Docker.md"
id="raw-url" group_item="true" data-view-component="true" class="js-permalink-replaceable-
link Button--secondary Button--small Button">
 Raw


```

```
border-0">
```

```

Blame
<article class="markdown-body entry-content container-lg" itemprop="text"><h1 tabindex="-1"
dir="auto"><a id="user-content-ungraded-lab-first-look-at-tensorflow-serving-with-docker"

```

```
<div class="d-flex">
```

```

height="16" aria-hidden="true"><path d="m7.775 3.275 1.25-1.25a3.5 3.5 0 1 1 4.95 4.95l-2.5
2.5a3.5 3.5 0 0 1-4.95 0 .751.751 0 0 1 .018-1.042.751.751 0 0 1 1.042-.018 1.998 1.998 0 0
0 2.83 0l2.5-2.5a2.002 2.002 0 0 0-2.83-2.83l-1.25 1.25a.751.751 0 0 1-1.042-.018.751.751 0
0 1-.018-1.042Zm-4.69 9.64a1.998 1.998 0 0 0 2.83 0l1.25-1.25a.751.751 0 0 1

```

```

<button class="btn-octicon btn-octicon-danger disabled tooltiped tooltiped-nw"
disabled
 aria-label="You must be signed in to make or propose changes" type="button">
 <svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-trash">
<path d="M11 1.75V3h2.25a.75.75 0 0 1 0 1.5H2.75a.75.75 0 0 1 0-1.5H5V1.75C5 .784 5.784 0
6.75 0h2.5C10.216 0 11 .784 11 1.75ZM4.496 6.675l.66 6.6a.25.25 0 0 0 .249.225h5.19a.25.25
0 0 0 .249-.225l.66-6.6a.75.75 0 0 1 1.492.149l-.66 6.6A1.748 1.748 0 0 1 10.595 15h-
5.19a1.75 1.75 0 0 1-1.741-1.575l-.66-6.6a.75.75 0 1 1 1.492-.15ZM6.5 1.75V3h3V1.75a.25.25
0 0 0-.25-.25h-2.5a.25.25 0 0 0-.25.25Z"></path>

```

this official TF [tutorial](#).

If you are a Windows user remember that **this lab is meant to be run using a WSL2 shell**. To open such shell use the Windows search bar and type either `wsl` or `bash`, one of these should be available if you installed WSL2 previously.

Open your terminal (or shell) and let's get started!

---

## ■ [Pulling the image](#)

---

Begin by pulling the TFS docker image from the Docker hub:

```
docker pull tensorflow/serving
```

This is the most minimal image that you can run TFS on. It contains all of the necessary dependencies to run TFS and was created with image size in mind, as a result of this it is around 400 mb in size.

---

## ■ [Clone the repo with the dummy model](#)

---

Now you will clone the official TFS [repo](#), which contains a dummy model named `Half Plus Two` that returns  $0.5 * x + 2$  for any value of `x`.

You will do the cloning in the temporary directory of your filesystem so that your machine does not get cluttered.

Run the following three commands:

```
https://github.com/tensorflow/serving">
```

```
mkdir -p /tmp/tfserving

cd /tmp/tfserving

git clone https://github.com/tensorflow/serving
```

These commands perform these operations in order:

- Create a directory called `tfserving` under the temporary directory `/tmp`.
- Change your current directory to the one that was just created
- Clone the repo in that location

After running these commands you can return to your previous directory by using `cd -` or you can simply close this command line window.

## ■ [Run Tensorflow Serving](#)

Since the image you just pulled contains all of the software needed to run the model under TFS, all that is left is to run a container out of the image.

A vanilla `docker run` looks like this:

```
docker run name-of-the-image
```

However, you can specify different flags to achieve different functionalities. You will see how this works in a bit.

Take a look at the command that will spin up a container to serve the model under TFS:

```
docker run --rm -p 8501:8501 \
 --mount type=bind,\
 source=/tmp/tfserving/serving/tensorflow_serving/servables/tensorflow/testdata/saved_model_half_plus_two_cpu,\
 target=/models/half_plus_two \
 -e MODEL_NAME=half_plus_two -t tensorflow/serving &
```

Wow, there is a lot of information in this command. Let's break it down to understand what every flag is doing:

- `--rm`: Delete this container after stopping running it. This is to avoid having to manually delete the container. Deleting unused containers helps your system to stay clean and tidy.
- `-p 8501:8501`: This flag performs an operation known as **port mapping**. The container, as well as your local machine, has its own set of ports. In order to access the `port 8501` within the container, you need to **map** it to a port on your computer. In this case it is mapped to the `port 8501` in your machine. This port is chosen as it is the default port to interact with the model through a `REST API`. If you were using a different protocol such as `gRPC` you will need to use `port 8500`. More information on this in the [tutorial](#) mentioned at the beginning of the lab.
- `--mount type=bind,source=dir/in/your/pc,target=dir/in/container`: This flag allows you to **mount** a directory in your pc to a directory within the container. This is very important because containers usually have short lifetimes and without mounting files onto them there is no way of persisting changes done to these files when the container was running.
- `-e MODEL_NAME=half_plus_two`: Will create the environment variable `MODEL_NAME` and assign to it the value of `half_plus_two`.
- `-t`: Attaches a pseudo-terminal to the container so you can check what is being printed in the standard streams of the container. This will allow you to see the logs printed out by TFS.

After running this command TFS will spin up and host the `Half Plus Two` model.

## ■ Consuming the model

Now that the model is being served on `port 8501` you can use an HTTP client to get predictions from it. Going forward you will be shown how to do this with `curl` but feel free to use any client of your choice.

Since you need to provide some data that the server will process you should use a `HTTP POST` request.

Let's do inference for a batch of three numbers, open a new command line window or tab and run the following command:

```
curl -d '{"instances": [1.0, 2.0, 5.0]}' \
-X POST http://localhost:8501/v1/models/half_plus_two:predict
```

As with the `docker run` command let's break down the flags in this one:

- `-d`: The `d` stands for data. This is the data that you are going to send to the server for it to process. Since you are communicating with the model via `REST` you should provide the data in a `JSON` format. TFS has the convention that the **key** of this object should be the string `instances` and the **value** should be a list that contains each data point that you want to make inference for.
- `-X`: This flag allows you to specify the desired `HTTP method`. By default `curl` uses the `GET` method but in this case it should be `POST`.

The last parameter of the command is the `URL` where it should make the request to. Let's break down the `URL` as well:

- `http://localhost:8501`: Stands for your own machine in the port 8501, as you specified earlier.
- `v1`: Refers to the version of TFS used. Right now this part of the `URL` will always be `v1`.
- `models/half_plus_two`: This part refers to what model should be served. Since you set the environment variable `MODEL_NAME` to have the value `half_plus_two`, this is the name of the model.
- `predict`: Allows TFS to know that the model is gonna be used for inference (or prediction).

After running the request you should be prompted with the prediction for each one of the three numbers you submitted.

## ■ Stopping the server

Finally you will learn how to stop the server running within the Docker container. In this case to stop the container is equivalent to stopping the server, to do so, run the following command to see all of the running Docker processes (or containers):

```
docker ps
```

This will display some relevant information for each running docker process. If you want to also check this information for stopped containers use the flag `-a`.



Docker automatically assigns a unique name to each container and this can be seen with the above command which should yield an output similar to this:

```
container_name
```

To stop a container simply use the command:

```
docker stop container_name
```

In this case the command will be:

```
docker stop laughing_yonath
```

After some seconds you should see that the process exited on the terminal that you spined up the container.

### Congratulations on finishing this ungraded lab!

Now you should have a better sense of how Docker can be leveraged to serve your Machine Learning models. You should also have a better understanding of how to use the `docker run` command to spin up containers and of how to use `curl` to interact with web servers.

This lab used a dummy model but in future labs you will see this process with real models and further TFS features.

### Keep it up!

```
</div>
```



```
</main>
```

```
<footer class="footer width-full container-xl p-responsive" role="contentinfo">
```

### ■ Footer



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```

<nav aria-label='Footer' class="col-12 col-lg-8">
 <h3 class='sr-only' id='sr-footer-heading'>Footer navigation</h3>
 <ul class="list-style-none d-flex flex-wrap col-12 flex-justify-center flex-lg-justify-between mb-2 mb-lg-0" aria-labelledby='sr-footer-heading'>
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 <li class="mr-3 mr-lg-0">Privacy
 <li class="mr-3 mr-lg-0"><a data-analytics-event="{"category":"Footer", "action":"go to security", "label":"text:security"}" href="https://github.com/security">Security
 <li class="mr-3 mr-lg-0">Status
 <li class="mr-3 mr-lg-0"><a data-ga-click="Footer, go to help, text:Docs" href="https://docs.github.com">Docs
 <li class="mr-3 mr-lg-0">Contact GitHub
 <li class="mr-3 mr-lg-0">Pricing
 <li class="mr-3 mr-lg-0">API
 <li class="mr-3 mr-lg-0">Training
 <li class="mr-3 mr-lg-0">Blog
 <a data-ga-click="Footer, go to about, text:about" href="https://github.com/about">About

</nav>

```

```
<template id="site-details-dialog">
```



```

<div class="Popover js-hovercard-content position-absolute" style="display: none; outline: none;" tabindex="0">

```

```
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```



```
</div>
```

```
<div id="js-global-screen-reader-notice" class="sr-only" aria-live="polite" ></div>
```